

Online PAC Labeling from a Linear Programming Perspective

STATS 357 Project

June 2, 2026

Abstract

We revisit cost-aware probably approximately correct (PAC) labeling through a linear-programming perspective. For a finite target pool, multiple cheap labeling sources, and access to an expert labeler, the oracle problem is to minimize labeling cost subject to an average-error constraint. The Lagrange dual of this LP gives a scalar shadow price for error, and the resulting itemwise minimizer is a fixed-price routing rule over cheap sources and expert deferral. Since the oracle losses are unavailable before labeling, the implementable method uses proxy losses only to generate candidate prices and policies. Finite-sample validity is then supplied by held-out calibration labels, which certify the actual loss of the selected policy without requiring the proxy to be correct. This separates optimization from inference: the LP and proxy model seek cost efficiency, while calibration provides the PAC certificate. We also describe an audited online extension in which randomized expert audits produce inverse-probability feedback for updating the dual price and controlling finite-horizon realized error.

1 Problem Setup and LP Formulation

1.1 Setup and Notation

There is a fixed pool of unlabeled items $X_1, \dots, X_T$, whose true labels $Y_1, \dots, Y_T$ are initially unknown, and the goal is to labeling the items correctly while minimizing labeling cost.

For each item t , we have access to k cheap ML models, which produce predicted labels $\hat{Y}_t^{(j)}$. Also, the expert label Y_t is treated as correct but incurs an expansive query cost $c_{t,E} > 0$. In this project, we assume ML models are much cheaper than the expert, so we set $c_{t,j} = 0$ for all t and $j \in [k]$ and $c_{t,E} = 1$. The labeling loss is bounded:

$$0 \leq \ell(Y_t, \hat{Y}_t^{(j)}) \leq 1, \quad t \in [T], j \in [k]. \quad (1)$$

The expert action has zero labeling loss. Define the action set $\mathcal{A} = \{E, 1, \dots, k\}$, where E denotes deferral to the expert. For $a \in \mathcal{A}$, define the loss and cost for item t by

$$h_t(a) = \begin{cases} 0, & a = E, \\ \ell(Y_t, \hat{Y}_t^{(a)}), & a \in [k], \end{cases} \quad c_t(a) = \begin{cases} 1, & a = E, \\ 0, & a \in [k]. \end{cases} \quad (2)$$

A labeling rule chooses an action $a_t \in \mathcal{A}$ for each item. The resulting final label is

$$\tilde{Y}_t = \begin{cases} Y_t, & a_t = E, \\ \hat{Y}_t^{(a_t)}, & a_t \in [k]. \end{cases}$$

1.2 The Cost-Aware Labeling LP

We formulate the cost-aware PAC labeling problem as a linear program. Let $x_{t,a}$ be the probability of choosing action a on item t . If the true losses $h_t(a)$ were known, the optimal labeling policy would be the solution to the following LP:

$$\begin{aligned}
& \text{minimize}_x && \frac{1}{T} \sum_{t=1}^T \sum_{a \in \mathcal{A}} c_t(a) x_{t,a} \\
& \text{subject to} && \frac{1}{T} \sum_{t=1}^T \sum_{a \in \mathcal{A}} h_t(a) x_{t,a} \leq \varepsilon, \\
& && \sum_{a \in \mathcal{A}} x_{t,a} = 1, \quad t \in [T], \\
& && x_{t,a} \geq 0, \quad t \in [T], a \in \mathcal{A}.
\end{aligned} \tag{3}$$

Intuitively, the LP minimizes labeling cost subject to an average-error constraint.

1.3 Lagrangian Dual and Shadow Price

The oracle LP (3) has a single global average-error constraint. Let $\lambda \geq 0$ denote the Lagrange multiplier, or shadow price, of the labeling error constraint. The dual problem is the one-dimensional concave maximization problem (See Appendix A.1 for details):

$$\max_{\lambda \geq 0} D(\lambda) := -\lambda \varepsilon + \frac{1}{T} \sum_{t=1}^T \min_{a \in \mathcal{A}} \{c_t(a) + \lambda h_t(a)\}. \tag{4}$$

This is the source of the fixed-price rule: once a price λ is fixed, the dual minimization separates across items, and the action for item t is given by the following simple rule:

$$a_t(\lambda) \in \arg \min_{a \in \mathcal{A}} \{c_t(a) + \lambda h_t(a)\}. \tag{5}$$

The dual price λ balances the trade-off between cost and error.

1.4 Roadmap

The LP perspective characterizes the optimal labeling policy by using a dual price λ . In practice, the true losses $h_t(a)$ are unknown before expert labels are observed, so the implementable method uses proxy losses $\hat{h}_t(a)$ to generate candidate prices and policies. The finite-sample validity of the selected policy is then supplied by held-out calibration labels, which certify the actual loss of the selected policy without requiring the proxy to be correct. This separates optimization from inference: the LP and proxy model seek cost efficiency, while calibration provides the PAC certificate.

In the **offline setting**, we use proxy losses $\hat{h}_t$ to generate fixed-price policies and reserve a calibration sample with expert labels to choose λ with a high-probability error certificate. The proxy $\hat{h}_t$ is used for efficient routing, while the calibration labels provide validity.

In the **online setting**, items arrive online and labels are revealed only intermittently through random audits. Whenever the true label is observed, we obtain feedback about the realized loss h_t and update the price λ_t through an online dual update.

2 Offline PAC Labeling

In this section, we first define the proxy-price labeling policies and then use a labeled calibration set to select a certified price for the target pool.

2.1 Price Estimation on A Calibration Set

We sample a calibration set $I_1, \dots, I_m$, define the empirical labeling loss

$$\hat{L}_m(\lambda) = \frac{1}{m} \sum_{s=1}^m H_{I_s}(\lambda). \quad (6)$$

Because $\hat{L}_m(\lambda)$ is monotone in λ , certifying the entire family costs no more than certifying a single price. We use the Hoeffding bound to give a level- $(1 - \alpha)$ upper confidence bound on $L_T(\lambda)$,

$$U_m(\lambda; \alpha) = \hat{L}_m(\lambda) + \sqrt{\frac{\log(1/\alpha)}{2m}}. \quad (7)$$

Then the price can be estimated by solving the offline LP dual problem with the confidence upper bound $U_m(\lambda; \alpha)$ in place of the true loss $L_T(\lambda)$:

$$\hat{\lambda} = \min \{ \lambda \geq 0 : U_m(\lambda; \alpha) \leq \varepsilon \}, \quad (8)$$

falling back to the expert-only policy $\lambda = \infty$ if no finite price qualifies. Because $C_T(\lambda)$ is non-decreasing and $U_m(\lambda; \alpha)$ non-increasing in λ , (8) coincides with the cost-minimizing selection $\arg \min_{\lambda} \{ C_T(\lambda) : U_m(\lambda; \alpha) \leq \varepsilon \}$. **WY: How to solve this problem? and here is a constraint on the size of calibration set.**

2.2 Labeling with Proxy Losses

For a candidate price λ , the oracle action $a_t(\lambda)$ from (5) cannot be computed because $h_t(a)$ is unknown. We therefore choose actions by replacing $h_t(a)$ with a proxy loss $\hat{h}_t(a)$.

The proxy may be obtained from a labeled training split by regression, isotonic calibration, multicalibration, or any other supervised procedure. The certification theorem below does not require $\hat{h}$ to be conservative or well calibrated. Better proxies reduce cost; held-out calibration supplies validity.

Given proxy $\hat{h}_t(a)$, the implementable rule is the plug-in version of (5):

$$a_t^\lambda \in \arg \min_{a \in \mathcal{A}} \{ c_t(a) + \lambda \hat{h}_t(a) \}.$$

Equivalently, the update rule is to first choose the model with the lowest proxy loss $j_t^\lambda \in \arg \min_{j \in [k]} \{ \hat{h}_{t,j} \}$, then defer to expert if and only if $\hat{h}_{t,j_t^\lambda} \geq 1/\lambda$. Denote $D_t^\lambda = \mathbf{1}\{ \hat{h}_{t,j_t^\lambda} \geq 1/\lambda \}$, then the action is

$$a_t^\lambda = \begin{cases} E, & D_t^\lambda = 1, \\ j_t^\lambda, & D_t^\lambda = 0. \end{cases} \quad (9)$$

The labeling loss and the cost given a price λ are then

$$H_t(\lambda) = (1 - D_t^\lambda) \ell(Y_t, \hat{Y}_t^{(j_t^\lambda)}), \quad C_t(\lambda) = D_t^\lambda. \quad (10)$$

Algorithm 1 Offline fixed-price labeling

Require: Unlabeled pool $X_1, \dots, X_T$; cheap predictions $\{\hat{Y}_t^{(j)}\}_{j \in [k]}$; proxy losses $\{\hat{h}_{t,j}\}_{j \in [k]}$; target error ε ; failure level α .

Stage 1: Price estimation on a calibration set

- 1: Sample calibration indices $I_1, \dots, I_m \sim \text{Unif}([T])$ and collect expert labels $\{Y_{I_s}\}_{s=1}^m$.
- 2: Form the empirical loss $\hat{L}_m(\lambda)$ (6) and the level- $(1 - \alpha)$ upper confidence bound $U_m(\lambda; \alpha)$ (7).
- 3: Select the certified price $\hat{\lambda} = \min\{\lambda \geq 0 : U_m(\lambda; \alpha) \leq \varepsilon\}$ (8).

Stage 2: Label the pool at price $\hat{\lambda}$

- 4: **for** $t = 1, \dots, T$ **do**
- 5: Pick the best cheap model $j_t \in \arg \min_{j \in [k]} \hat{h}_{t,j}$.
- 6: **If** $\hat{h}_{t,j_t} \geq 1/\hat{\lambda}$: query expert, $\tilde{Y}_t \leftarrow Y_t$.
- 7: **Else**: accept cheap label, $\tilde{Y}_t \leftarrow \hat{Y}_t^{(j_t)}$.
- 8: **end for**

Ensure: Labels $\tilde{Y}_1, \dots, \tilde{Y}_T$ with certified error $\leq \varepsilon$ at level $1 - \alpha$.

Then the average loss and cost of the price- λ policy are

$$L_T(\lambda) = \frac{1}{T} \sum_{t=1}^T H_t(\lambda), \quad C_T(\lambda) = \frac{1}{T} \sum_{t=1}^T C_t(\lambda). \quad (11)$$

The overall offline labeling algorithm is summarized in Algorithm 1.

Relation to PAC labeling Under the single-source case, if we use uncertainty score as the proxy, $\hat{h}_t = U_t$, then AI label is used when

$$\hat{h}_t = U_t < 1/\hat{\lambda}.$$

This corresponds to the threshold policy used in the original PAC labeling method [1]. Differently, the threshold used in our method is defined by the dual price motivated from the LP perspective. With multiple sources, $\hat{h}_t$ is replaced by the itemwise minimum of the proxy losses.

2.3 Offline PAC Guarantees

The transductive setting is closest to fixed-dataset PAC labeling. The pool $X_1, \dots, X_T$ is fixed, and calibration indices $I_1, \dots, I_m$ are drawn independently from $\text{Unif}([T])$. The expert labels Y_{I_s} are then revealed. For each fixed λ ,

$$\mathbb{E}[H_{I_s}(\lambda) \mid X_{1:T}, Y_{1:T}] = L_T(\lambda).$$

Theorem 1 (Offline PAC guarantee, monotone single-price form). *Assume (1) and equal cheap-source costs, so that (??) holds and $L_T(\lambda)$ is non-increasing in λ . Let the proxy $\hat{h}$ —hence the score $\hat{h}_t^*$ and the nested family $\{\Pi_\lambda\}$ —be constructed independently of the calibration labels, conditional on the target pool, cheap-source predictions, and any separate training data. Suppose that for each fixed price the single-price bound (7) is a valid level- $(1 - \alpha)$ upper confidence bound on $L_T(\lambda)$, required one λ at a time, not simultaneously. Let $\hat{\lambda}$ be selected by (8). Then, with probability at least $1 - \alpha$ over the random draw of the calibration indices,*

$$L_T(\hat{\lambda}) \leq \varepsilon. \quad (12)$$

Consequently the labels produced by the price- $\hat{\lambda}$ policy $a_t^{\hat{\lambda}}$ satisfy

$$\frac{1}{T} \sum_{t=1}^T \ell(Y_t, \tilde{Y}_t) \leq \varepsilon \quad (13)$$

with probability at least $1 - \alpha$; if calibration items are corrected to expert labels in the final dataset, the actual final loss is no larger than $L_T(\hat{\lambda})$.

The proof is in Appendix A. The key point is monotonicity, not uniformity. Although the algorithm selects among many candidate prices, $L_T(\lambda)$ is non-increasing in λ by (??), so the only way the selected price can be infeasible is for the bound to fail at the single deterministic price where the true loss last crosses ε . A union bound over Λ is therefore unnecessary, exactly as in single-source PAC labeling and in the Dvoretzky–Kiefer–Wolfowitz inequality, whose freedom from multiplicity corrections also rests on monotonicity.

Corollary 1 (Inductive deployment guarantee). *Suppose the calibration set is an independent labeled sample from a distribution P , and the deployment items are drawn independently from the same distribution. For a price λ , define the population risk*

$$L(\lambda) = \mathbb{E}_{(X,Y) \sim P} \left[(1 - D^\lambda(X)) \ell(Y, \hat{Y}^{(j^\lambda(X))}(X)) \right].$$

Assume Λ is finite and independent of the calibration labels conditional on covariates, cheap-source predictions, costs, and any separate training data used to construct $\hat{h}$. Let $\hat{\lambda}$ be selected by (8) using the calibration sample. Then $L(\hat{\lambda}) \leq \varepsilon$ with probability at least $1 - \alpha$. For a finite deployment pool of size T , with probability at least $1 - \alpha - \delta$,

$$\frac{1}{T} \sum_{t=1}^T \ell(Y_t, \tilde{Y}_t) \leq \varepsilon + \sqrt{\frac{\log(1/\delta)}{2T}}. \quad (14)$$

If the desired realized deployment-pool average is at most ε , the internal certification target should be reduced by the final square-root term.

3 Online PAC Labeling via Audits

The offline guarantee certifies a fixed policy after a labeled calibration stage. In an online data-curation problem, inputs arrive sequentially, decisions are made before future data are seen, and the loss of an AI-labeled item is unobserved unless an expert label is revealed. Therefore the offline calibration argument cannot be reused directly. The natural online analogue is to keep the dual-price decision rule but add randomized audits of AI-labeled points.

3.1 Online Streaming Setup and Audited Feedback

At time t , the learner observes X_t , cheap labels $\hat{Y}_t^{(j)}$, costs $c_{t,j}$, expert cost $c_{t,E}$, and predictable proxy losses $\hat{h}_{t,j}$. The price λ_t is chosen from past information. The online fixed-price decision is

$$j_t^* \in \arg \min_{j \in [k]} \{c_{t,j} + \lambda_t \hat{h}_{t,j}\}, \quad (15)$$

followed by expert deferral if

$$c_{t,E} \leq c_{t,j_t^*} + \lambda_t \hat{h}_{t,j_t^*}. \quad (16)$$

Let $D_t = 1$ denote this final-label deferral decision. The quantities j_t^* , D_t , and the loss H_t defined below are the online counterparts of the offline j_t^λ , D_t^λ , and $H_t(\lambda)$ from Section 2.2, evaluated at the time-varying price λ_t ; we drop the explicit price superscript since it is implied by the round index t . If $D_t = 1$, the output label is Y_t and the final loss is zero. If $D_t = 0$, the learner initially outputs $\hat{Y}_t^{(j_t^*)}$. It then draws an independent audit variable

$$A_t \sim \text{Bernoulli}(p_t), \quad p_t \in [p_{\min}, 1].$$

If $A_t = 1$, the expert label is revealed and the final released label may be corrected to Y_t . The audit still records the counterfactual AI error for updating the price.

Define the counterfactual AI-decision loss

$$H_t = (1 - D_t)\ell(Y_t, \hat{Y}_t^{(j_t^*)}),$$

where deferral has zero loss because the expert label is used. Deferral is an observed zero-loss event, while an unaudited AI decision has unobserved loss. Define the augmented observation mask and its conditional observation probability by

$$O_t = D_t + (1 - D_t)A_t, \quad q_t = D_t + (1 - D_t)p_t. \quad (17)$$

Thus $q_t = 1$ when $D_t = 1$ and $q_t = p_t$ when $D_t = 0$. The centered inverse-observation update signal is

$$\hat{Z}_t = \frac{O_t}{q_t} (H_t - \varepsilon_0). \quad (18)$$

The realized final loss after audit-and-correct is

$$G_t = \ell(Y_t, \tilde{Y}_t), \quad G_t \leq H_t.$$

The online price update is

$$\lambda_{t+1} = [\lambda_t + \eta \hat{Z}_t]_+, \quad (19)$$

where $\eta > 0$ is a step size and $\varepsilon_0 \leq \varepsilon$ is an internal error target.

3.2 Online Finite-Horizon Guarantee

The online result is distribution-free with respect to the data sequence; the probability is over the algorithm's audit randomness. It controls realized error, not cost regret against the offline LP optimum.

Assumption 1 (Predictable routing and random audits). *The routing decision, audit probability, and selected source are chosen using only past information and the current non-expert information $(X_t, \{\hat{Y}_t^{(j)}, \hat{h}_{t,j}, c_{t,j}\}_{j=1}^k, c_{t,E})$. After this decision, the audit draw is independent Bernoulli with known probability $p_t \in [p_{\min}, 1]$.*

Assumption 2 (Proxy floor and bounded expert cost). *For some $\rho > 0$, $\hat{h}_{t,j} \in [\rho, 1]$ for all t, j . For the fixed horizon, let*

$$c_{\min} = \min_{t \leq T, j \in [k]} c_{t,j}, \quad c_E^{\max} = \max_{t \leq T} c_{t,E}.$$

If future costs are not known in advance, these extrema may be replaced by any deterministic lower bound on cheap-source costs and upper bound on expert costs that are known before deployment.

Algorithm 2 Audited online dual-price PAC labeling

Require: Stream horizon T ; target error ε ; audit floor $p_{\min}$; step size η ; internal target ε_0 .

```
1: Initialize  $\lambda_1 \geq 0$ .
2: for  $t = 1, \dots, T$  do
3:   Observe  $X_t$ , cheap labels  $\{\hat{Y}_t^{(j)}\}_{j=1}^k$ , costs, and predictable proxies  $\{\hat{h}_{t,j}\}_{j=1}^k$ .
4:   Choose  $j_t^*$  by (15).
5:   if (16) holds then
6:     Query the expert, output  $\tilde{Y}_t = Y_t$ , set  $D_t = 1$  and  $\hat{Z}_t = -\varepsilon_0$ .
7:   else
8:     Output the AI label  $\tilde{Y}_t = \hat{Y}_t^{(j_t^*)}$  and set  $D_t = 0$ .
9:     Draw  $A_t \sim \text{Bernoulli}(p_t)$  with  $p_t \in [p_{\min}, 1]$ .
10:    if  $A_t = 1$  then
11:      Query the expert, optionally correct  $\tilde{Y}_t = Y_t$ , and set  $\hat{Z}_t = \{\ell(Y_t, \hat{Y}_t^{(j_t^*)}) - \varepsilon_0\}/p_t$ .
12:    else
13:      Set  $\hat{Z}_t = 0$ .
14:    end if
15:  end if
16:  Update  $\lambda_{t+1}$  by (19).
17: end for
```

Proposition 1 (Unbiased intermittent update signal). *Under bounded loss and online predictability,*

$$\mathbb{E} \left[\frac{O_t}{q_t} \middle| \mathcal{G}_t \right] = 1, \quad \mathbb{E}[\hat{Z}_t \mid \mathcal{G}_t] = H_t - \varepsilon_0.$$

Theorem 2 (Audited online finite-horizon control). *Assume bounded loss, online predictability and audits, and the proxy floor. Let*

$$\lambda_{\text{def}} = \frac{(c_E^{\max} - c_{\min})_+}{\rho}, \quad \Lambda_{\text{on}} = \max \left\{ \lambda_1, \lambda_{\text{def}} + \frac{\eta(1 - \varepsilon_0)}{p_{\min}} \right\}.$$

Run Algorithm 2 with constant step size $\eta > 0$ and internal target $\varepsilon_0 \in [0, \varepsilon]$. Then, for any fixed horizon T and any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the audit randomness,

$$\frac{1}{T} \sum_{t=1}^T H_t \leq \varepsilon_0 + \frac{\Lambda_{\text{on}}}{\eta T} + \frac{1}{p_{\min}} \sqrt{\frac{2 \log(1/\delta)}{T}}. \quad (20)$$

Consequently, if audited AI labels are corrected to expert labels in the final dataset, then

$$\frac{1}{T} \sum_{t=1}^T G_t \leq \varepsilon_0 + \frac{\Lambda_{\text{on}}}{\eta T} + \frac{1}{p_{\min}} \sqrt{\frac{2 \log(1/\delta)}{T}}. \quad (21)$$

Corollary 2 (PAC-style online tuning). *Under the conditions of Theorem 2, suppose*

$$\varepsilon_0 + \frac{\Lambda_{\text{on}}}{\eta T} + \frac{1}{p_{\min}} \sqrt{\frac{2 \log(1/\delta)}{T}} \leq \varepsilon,$$

with $\varepsilon_0 \geq 0$. Then the audit-and-correct final labels satisfy

$$\frac{1}{T} \sum_{t=1}^T G_t \leq \varepsilon$$

with probability at least $1 - \delta$. With $\eta \asymp T^{-1/2}$ and constant $p_{\min}$, the slack is $O(T^{-1/2})$.

3.3 Offline versus Online Certification

The online theorem is deliberately different from the offline one. Offline calibration certifies a fixed candidate policy by evaluating it on a held-out labeled sample. Online auditing controls the long-run error of a moving policy by converting unobserved AI losses into an unbiased intermittent-feedback signal. The proxy losses again affect efficiency rather than validity, but in the online setting some audit or reveal mechanism is essential: without it, two streams can induce identical observations while having opposite AI-label correctness on every unaudited item.

4 Experiments

We run a small proof-of-concept study on ImageNet and ImageNetV2 using the confidence score already included in the data. There is one cheap source, namely the model prediction, and one expert source, namely the ground-truth label. The cheap-source proxy loss is $\hat{h} = 1 - \text{confidence}$, the cheap source has zero marginal cost, and the expert has unit cost. All reported numbers average over 500 random trials. Appendix B gives the full sweep and implementation details.

Offline calibration. The offline experiment compares the fixed-price rule from Section 2 with the original scalar PAC threshold baseline. Because this experiment has only one cheap source, the fixed-price policies are nested uncertainty-threshold rules. We therefore certify the fixed-price rule with the sharper monotone confidence bound

$$\hat{L}_m(\lambda) + \sqrt{\frac{\log(1/\alpha)}{2m}},$$

rather than the finite-class union bound used for general multi-source policy classes. At $\varepsilon = 0.10$, both methods have zero violations in 500 trials and nearly identical expert-budget savings (Table 1). Figure 1 shows the same comparison across all three ε levels, and Table 4 gives the corresponding full sweep.

Dataset	Method	Error	Violation	Budget saved
ImageNet	fixed-price monotone UCB	0.0873	0.0000	0.7848
ImageNet	original PAC	0.0879	0.0000	0.7864
ImageNetV2	fixed-price monotone UCB	0.0718	0.0000	0.5457
ImageNetV2	original PAC	0.0726	0.0000	0.5479

Table 1: Offline results at $\varepsilon = 0.10$. “Budget saved” is one minus the fraction of items sent to the expert.

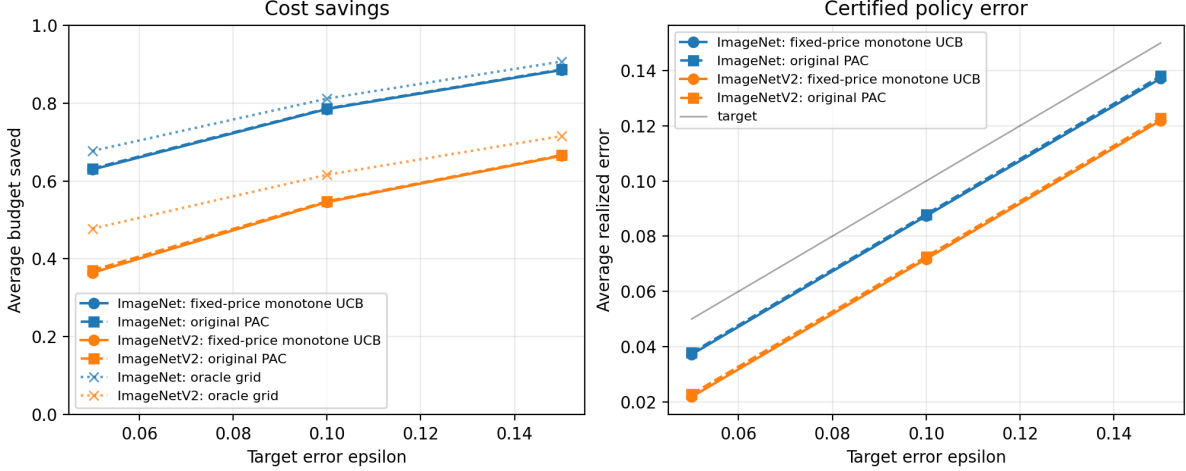


Figure 1: Offline sweep over target error levels. After exploiting the single-source monotonicity, the fixed-price rule is nearly indistinguishable from the original scalar PAC baseline; the remaining gap comes from using a fixed threshold grid rather than the exact calibration-order threshold.

Two-source synthetic routing. The single-source experiment above is the special case in which the LP and the original scalar PAC threshold coincide. To stress-test the multi-source regime that motivates the LP formulation, we synthesize two complementary cheap sources from each base ImageNet table. Items are partitioned by the parity of the true class label. On items inside its specialty each source reuses the original predictor’s label and confidence; on items outside its specialty each source emits a uniform-random label and a confidence drawn from $\text{Uniform}[0, 0.2]$, so the source is calibrated to its own competence. Source A’s specialty is the even-class half and source B’s is the odd-class half. Neither source dominates; standalone accuracy is approximately 0.39 on ImageNet and 0.32 on ImageNetV2, while the per-item oracle that picks whichever source is correct attains 0.78 and 0.65, respectively. Costs match the simple offline experiment: $c_A = c_B = 0$, $c_E = 1$, so the dual rule routes to $\arg \min_j \hat{h}_{t,j}$ and defers when $\min_j \hat{h}_{t,j} > 1/\lambda$. We compare three calibrated policies. *LP two-source uniform UCB* is the fixed-price rule of Section 2.1 applied to the constructed two-source pool, certified with the conservative finite-class Hoeffding bound (??) over a $|\Lambda| = 202$ threshold grid; we report it as a worst-case baseline. We emphasize, however, that with $c_A = c_B = 0$ the routed loss *is* monotone in the scalar score $\hat{h}_t^* = \min_j \hat{h}_{t,j}$, because the source map $\arg \min_j \hat{h}_{t,j}$ does not depend on λ , so Theorem 1 licenses the sharper single-price bound (7) here as well; this replaces the $\sqrt{\log(|\Lambda|/\alpha)}$ radius by $\sqrt{\log(1/\alpha)}$ and is most consequential in the tight- ε regime discussed below. *Source- j monotone PAC* ($j \in \{A, B\}$) is the original PAC baseline restricted to one source. *Best-of-singles (Bonferroni)* runs both single-source baselines at level $\alpha/2$ and picks the source with the larger empirical savings on the calibration sample, which remains a valid α -PAC procedure.

At $\varepsilon = 0.10$ all four methods have zero violations in 500 trials. The LP rule saves roughly 0.29 more of the expert budget on ImageNet and 0.13 more on ImageNetV2 than the best Bonferroni-corrected single-source baseline (Table 2). The gap on ImageNet is close to the LP oracle’s envelope (0.81 saved at $\varepsilon = 0.10$), confirming that the per-item routing benefit dominates the finite-policy union-bound radius once the calibration sample is large enough. At $\varepsilon = 0.05$ on ImageNetV2 the order reverses: with $m = 2000$ the uniform radius $\sqrt{\log(202/\alpha)/(2m)} \approx 0.046$ leaves almost no headroom under $\varepsilon = 0.05$, so the LP rule certifies near-zero savings while each single-source

Dataset	Method	Error	Violation	Budget saved
ImageNet	LP two-source uniform UCB	0.0793	0.0000	0.7645
ImageNet	Best-of-singles (Bonferroni)	0.0879	0.0000	0.4746
ImageNet	Source A monotone PAC	0.0878	0.0000	0.4715
ImageNet	Source B monotone PAC	0.0877	0.0000	0.4754
ImageNetV2	LP two-source uniform UCB	0.0534	0.0000	0.4923
ImageNetV2	Best-of-singles (Bonferroni)	0.0720	0.0000	0.3587
ImageNetV2	Source A monotone PAC	0.0727	0.0000	0.3470
ImageNetV2	Source B monotone PAC	0.0730	0.0000	0.3629

Table 2: Two-source synthetic routing at $\varepsilon = 0.10$. The LP method routes each item to the source with the lower proxy loss and certifies the routed average error against the union bound over the 202-threshold grid. The largest budget saved in each dataset is shown in bold.

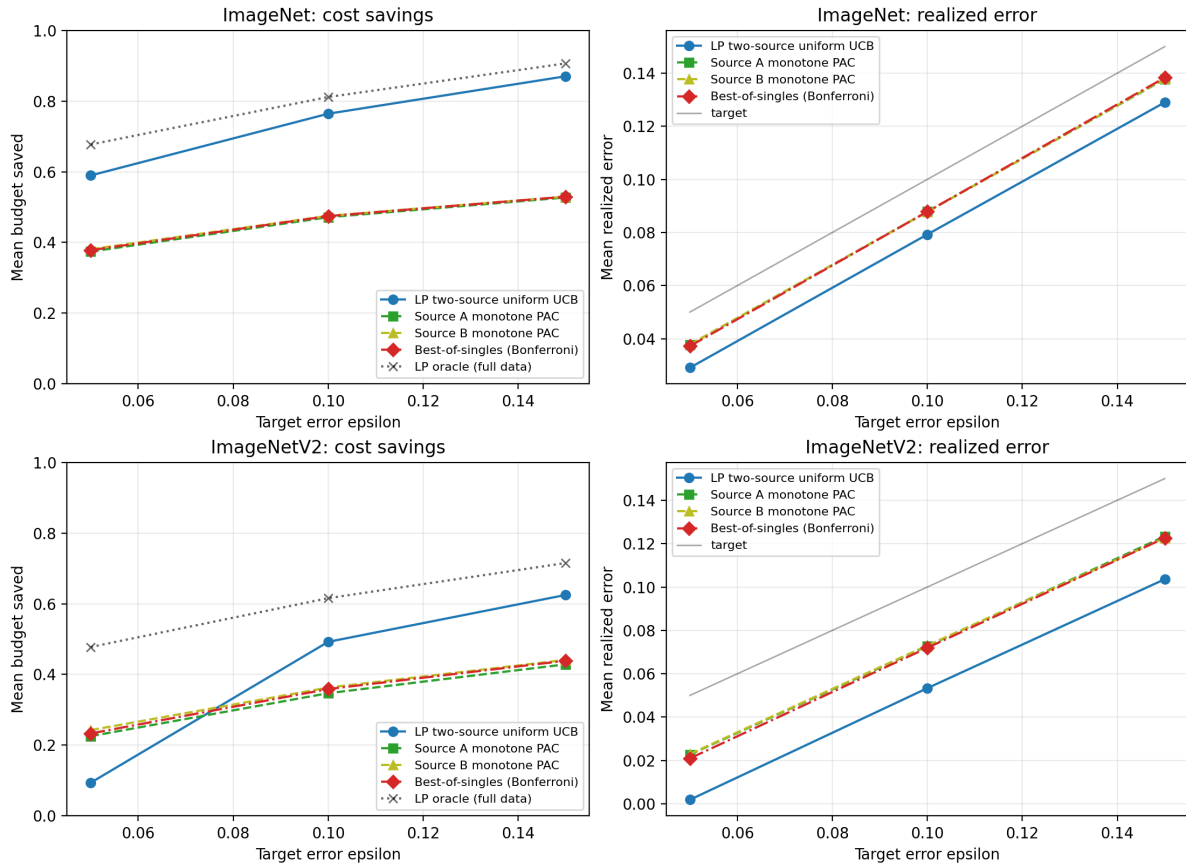


Figure 2: Two-source synthetic routing sweep. Each single source can certify at most its own specialty half, so the three single-source curves saturate near 0.55 (ImageNet) and 0.45 (ImageNetV2). The LP method recovers most of the per-item oracle envelope by routing across sources at the cost of a $\sqrt{\log|\Lambda|}$ uniform-certification radius. On ImageNetV2 at $\varepsilon = 0.05$ that radius approaches ε itself and the LP method is forced into near-full deferral, illustrating where a tighter sup-deviation bound would be needed.

baseline still certifies its specialty half (Figure 2; Table 5 in the appendix reports the full sweep). With equal cheap-source costs this gap is removed rather than merely shrunk: the routed loss is monotone in the unidimensional statistic $\hat{h}_t^* = \min_j \hat{h}_{t,j}$, so Theorem 1 certifies with the single-price radius $\sqrt{\log(1/\alpha)/(2m)}$ and pays no $\sqrt{\log|\Lambda|}$ penalty. The uniform-UCB numbers above are therefore a conservative baseline; the monotone certificate recovers the headroom lost at small ε . (A variance-adaptive Bernstein bound would tighten the constant further; we leave that to future work.)

Online audited routing. The online experiment uses a random stream order and the dual-price update in Section 3 with audit probabilities $p \in \{0.1, 0.2, 0.5\}$. The update follows the intermittent-feedback form: an unaudited AI decision does not move the price, while an expert deferral is treated as an observed zero-loss event. These runs set the internal online target to $\varepsilon_0 = \varepsilon$, so they are a behavioral check of the audited update rather than a theorem-tight slack-adjusted implementation of Corollary 2. Lower audit rates save more expert budget but leave less correction, so final error is closer to the target. At $\varepsilon = 0.10$, ImageNet has no final-error violations for all three audit rates; ImageNetV2 has a small violation rate at $p = 0.1$ and none at $p \geq 0.2$ (Table 3; the full sweep across $\varepsilon \in \{0.05, 0.10, 0.15\}$ is in Table 6).

Dataset	p	Final error	AI error	Violation	Budget saved
ImageNet	0.1	0.0780	0.0866	0.0000	0.6629
ImageNet	0.2	0.0737	0.0921	0.0000	0.6115
ImageNet	0.5	0.0490	0.0981	0.0000	0.3952
ImageNetV2	0.1	0.0863	0.0958	0.0400	0.5091
ImageNetV2	0.2	0.0783	0.0979	0.0000	0.4681
ImageNetV2	0.5	0.0499	0.0996	0.0000	0.3010

Table 3: Online audited results at $\varepsilon = 0.10$. “AI error” is the counterfactual average error of AI-labeled items before audit correction; “Final error” is after audited items are corrected by the expert.

5 Related Work

PAC labels and conformal calibration. The closest starting point is PAC labeling [1], which gives finite-sample guarantees for cost-effective dataset labeling using AI predictions and expert audits. That framework already includes both scalar-threshold PAC labeling and multi-model PAC routing variants. Our formulation is best viewed as a complementary optimization-first presentation: we start from a finite-pool cost-minimization LP, derive the fixed-price routing rule from its dual, allow item- and source-specific costs, and then use calibration only to certify the actual loss of the selected proxy-generated policy. The calibration logic is closely related to conformal prediction and distribution-free uncertainty quantification [2, 3], but the object being controlled here is the average error of a released labeled dataset rather than coverage of prediction sets.

Online conformal prediction with partial feedback. Adaptive and online conformal methods update calibration parameters over time to handle changing environments [4]. Recent work has studied intermittent or semi-bandit feedback, where labels are not always observed [5, 6, 7]. Those

papers motivate the audited online extension in Section 3: when labels are not always revealed, the feedback used to update a calibration or price parameter must be corrected for the observation process.

Learning to defer and cost-sensitive routing. Learning-to-defer methods train systems that either predict or pass an item to an expert [8, 9]. Multiple-expert and cost-sensitive variants study richer deferral and routing settings [10, 11]. The distinction here is the finite-sample PAC certificate for the final released labels. The proxy router may be learned by standard supervised methods, but the guarantee comes from held-out calibration of the actual selected policy.

6 Discussion

The report separates two regimes. The offline method has a clean division of labor. The proxy model and the LP structure make the policy cost efficient; the calibration set makes it valid. When instantiating the method, the main modeling choice is whether the target is transductive or inductive. If the application is dataset curation for a fixed unlabeled pool, the transductive theorem should be the main result. If the application is repeated deployment on future items, the inductive statement should be foregrounded. The online method addresses a different obstacle: the selected policy changes over time and AI losses are not observed without expert reveal. Random audits supply the missing feedback through an inverse-probability loss estimate. This gives finite-horizon error control, but not a cost-regret comparison to the offline oracle LP. The experiments in Section 4 illustrate this distinction: offline calibration gives a clean finite-policy certificate, while the online mechanism exposes an explicit audit-cost versus correction tradeoff. The main empirical limitation is that the current data contain only one cheap source. A stronger test of the LP perspective would use several cheap sources with heterogeneous costs, where the original scalar threshold structure no longer captures the routing problem.

A Proofs

A.1 Dual Derivation

Introduce one dual variable $\lambda \geq 0$ for the average-error constraint and one free dual variable $\mu_t \in \mathbb{R}$ for each simplex equality. The nonnegativity constraints $x_{t,a} \geq 0$ are kept as the domain of the inner minimization.

Primal constraint	Dual variable	Sign restriction
$\frac{1}{T} \sum_{t,a} h_t(a)x_{t,a} \leq \varepsilon$	λ	$\lambda \geq 0$
$\sum_a x_{t,a} = 1$ for item t	μ_t	$\mu_t \in \mathbb{R}$
$x_{t,a} \geq 0$	domain constraint	—

Using the equality convention $\mu_t(1 - \sum_a x_{t,a})$, the Lagrangian is

$$\begin{aligned}\mathcal{L}(x; \lambda, \mu) &= \frac{1}{T} \sum_{t,a} c_t(a) x_{t,a} + \lambda \left(\frac{1}{T} \sum_{t,a} h_t(a) x_{t,a} - \varepsilon \right) + \sum_{t=1}^T \mu_t \left(1 - \sum_a x_{t,a} \right) \\ &= \sum_{t,a} \left[\frac{1}{T} \{c_t(a) + \lambda h_t(a)\} - \mu_t \right] x_{t,a} + \sum_{t=1}^T \mu_t - \lambda \varepsilon.\end{aligned}\tag{22}$$

For fixed (λ, μ) , the inner problem is $\inf_{x \geq 0} \mathcal{L}(x; \lambda, \mu)$. Since the Lagrangian is linear in every $x_{t,a}$, this infimum is finite if and only if

$$\mu_t \leq \frac{1}{T} \{c_t(a) + \lambda h_t(a)\}, \quad t \in [T], \quad a \in \mathcal{A}.\tag{23}$$

Otherwise some variable with a negative coefficient can be sent to $+\infty$, driving the Lagrangian to $-\infty$. When (23) holds, the x terms vanish at the inner minimum and the dual function is

$$g(\lambda, \mu) = \sum_{t=1}^T \mu_t - \lambda \varepsilon.$$

Thus the explicit dual LP is

$$\begin{aligned}& \text{maximize}_{\lambda, \mu} \quad \sum_{t=1}^T \mu_t - \lambda \varepsilon \\ & \text{subject to} \quad \mu_t \leq \frac{1}{T} \{c_t(a) + \lambda h_t(a)\}, \quad t \in [T], \quad a \in \mathcal{A}, \\ & \quad \lambda \geq 0, \quad \mu_t \in \mathbb{R}, \quad t \in [T].\end{aligned}\tag{24}$$

For any fixed λ , the objective is increasing in each μ_t , so the optimal value is attained at

$$\mu_t^*(\lambda) = \frac{1}{T} \min_{a \in \mathcal{A}} \{c_t(a) + \lambda h_t(a)\}.$$

Substitution gives the reduced dual (4). The primal is feasible because choosing the expert for all items gives zero loss, and it is bounded because costs are nonnegative and finite. Therefore strong duality holds by finite-dimensional linear programming duality.

Proof of Theorem 1. Condition on the target pool and the proxy, so that $j_t^*, \hat{h}_t^*$, the nested family $\{\Pi_\lambda\}$, and the per-routed-item losses $\ell(Y_t, \hat{Y}_t^{(j_t^*)}) \in [0, 1]$ are all deterministic. By (??) a routed item contributes its fixed loss and a deferred item contributes zero, and the routed set $\{t : \hat{h}_t^* < 1/\lambda\}$ shrinks as λ grows; hence $H_t(\lambda)$ is non-increasing in λ for every t , and so are $L_T(\lambda)$ and $\hat{L}_m(\lambda)$. For each fixed λ the calibration draws $H_{I_s}(\lambda)$ are i.i.d. in $[0, 1]$ with mean $L_T(\lambda)$, so Hoeffding's inequality gives the single-price bound, valid one λ at a time,

$$\mathbb{P}\{L_T(\lambda) > U_m(\lambda; \alpha)\} \leq \alpha, \quad U_m(\lambda; \alpha) = \hat{L}_m(\lambda) + \sqrt{\frac{\log(1/\alpha)}{2m}}.$$

Let $\lambda^\circ = \max\{\lambda \in \Lambda : L_T(\lambda) > \varepsilon\}$ be the largest price whose true loss still exceeds the target; if no such price exists then even $\lambda = 0$ is feasible and the claim is immediate. By monotonicity every price strictly above λ° is feasible, so it suffices to show $\hat{\lambda} > \lambda^\circ$ with probability at least $1 - \alpha$. Suppose

instead $\hat{\lambda} \leq \lambda^\circ$. Since $\hat{\lambda}$ is the smallest price with $U_m(\lambda; \alpha) \leq \varepsilon$ and $U_m(\cdot; \alpha)$ is non-increasing, $U_m(\lambda^\circ; \alpha) \leq U_m(\hat{\lambda}; \alpha) \leq \varepsilon$; but $L_T(\lambda^\circ) > \varepsilon$ by definition, so the bound fails at the single fixed price λ° , an event of probability at most α by single-price validity. Hence $\mathbb{P}\{\hat{\lambda} > \lambda^\circ\} \geq 1 - \alpha$, and on that event $L_T(\hat{\lambda}) \leq \varepsilon$. Integrating over the proxy preserves the bound. The final label loss equals $H_t(\hat{\lambda})$ on items not corrected by the expert and is no larger on calibration items released with their expert labels. This proves the claim. $\square$

Proof of Corollary 1. For each fixed λ the calibration losses are i.i.d. in $[0, 1]$ with mean $L(\lambda)$, and under equal cheap-source costs $L(\lambda)$ is non-increasing in λ by the same argument as in Theorem 1. The single-price bound (7) is therefore valid one λ at a time, and the monotone single-point argument of that proof—applied to $L(\lambda)$ in place of $L_T(\lambda)$ —gives $L(\hat{\lambda}) \leq \varepsilon$ with probability at least $1 - \alpha$ over the calibration sample. Conditional on the calibration sample and on this event, the deployment losses $\ell(Y_t, \tilde{Y}_t)$ are independent, bounded in $[0, 1]$, and have conditional mean at most $L(\hat{\lambda}) \leq \varepsilon$. Hoeffding’s inequality therefore gives

$$\frac{1}{T} \sum_{t=1}^T \ell(Y_t, \tilde{Y}_t) \leq \varepsilon + \sqrt{\frac{\log(1/\delta)}{2T}}$$

with conditional probability at least $1 - \delta$. Combining the calibration and deployment events gives probability at least $1 - \alpha - \delta$. $\square$

Proof of Proposition 1. For the analysis, let $\mathcal{G}_t$ denote the post-decision sigma-field that fixes the selected source, audit probability, deferral decision, and the latent selected loss. The learner need not observe the latent loss; it is included only so the conditional expectation is well defined. If $D_t = 1$, then $O_t = q_t = 1$ and $H_t = 0$, so

$$\mathbb{E}\left[\frac{O_t}{q_t} \middle| \mathcal{G}_t\right] = 1, \quad \mathbb{E}[\hat{Z}_t \mid \mathcal{G}_t] = -\varepsilon_0 = H_t - \varepsilon_0.$$

If $D_t = 0$, then $O_t = A_t$, $q_t = p_t$, and conditional on $\mathcal{G}_t$ the audit draw is Bernoulli(p_t) while the loss $\ell(Y_t, \hat{Y}_t^{(j^*)})$ is fixed. Hence

$$\mathbb{E}\left[\frac{O_t}{q_t} \middle| \mathcal{G}_t\right] = \frac{p_t}{p_t} = 1.$$

Since H_t and ε_0 are fixed conditional on $\mathcal{G}_t$, this also gives

$$\mathbb{E}[\hat{Z}_t \mid \mathcal{G}_t] = \mathbb{E}\left[\frac{O_t}{q_t}(H_t - \varepsilon_0) \middle| \mathcal{G}_t\right] = H_t - \varepsilon_0.$$

$\square$

Proof of Theorem 2. First bound the price process. If $\lambda_t \geq \lambda_{\text{def}}$, then for every cheap source j ,

$$c_{t,j} + \lambda_t \hat{h}_{t,j} \geq c_{\min} + \lambda_{\text{def}} \rho \geq c_E^{\max} \geq c_{t,E}.$$

Thus the algorithm defers to the expert, so $D_t = 1$ and $\hat{Z}_t = -\varepsilon_0$. Consequently $\lambda_{t+1} = [\lambda_t - \eta \varepsilon_0]_+ \leq \lambda_t$. If instead $\lambda_t < \lambda_{\text{def}}$, then $\hat{Z}_t \leq (1 - \varepsilon_0)/p_{\min}$ and hence

$$\lambda_{t+1} \leq \lambda_t + \eta(1 - \varepsilon_0)/p_{\min} < \lambda_{\text{def}} + \eta(1 - \varepsilon_0)/p_{\min}.$$

By induction, $0 \leq \lambda_t \leq \Lambda_{\text{on}}$ for all t .

Since $[x]_+ \geq x$, the update gives

$$\lambda_{t+1} \geq \lambda_t + \eta \hat{Z}_t.$$

Summing over $t = 1, \dots, T$ yields

$$\sum_{t=1}^T \hat{Z}_t \leq \frac{\lambda_{T+1} - \lambda_1}{\eta} \leq \frac{\Lambda_{\text{on}}}{\eta}.$$

Let $M_t = (H_t - \varepsilon_0) - \hat{Z}_t$. By Proposition 1, M_t is a martingale difference with respect to the audit filtration. Moreover, $|M_t| \leq 1/p_{\min}$. Azuma-Hoeffding gives, with probability at least $1 - \delta$,

$$\sum_{t=1}^T M_t \leq \frac{1}{p_{\min}} \sqrt{2T \log(1/\delta)}.$$

Because

$$\sum_{t=1}^T H_t = T\varepsilon_0 + \sum_{t=1}^T \hat{Z}_t + \sum_{t=1}^T M_t,$$

combining the last displays gives (20). Finally, audit-and-correct gives $G_t \leq H_t$ pointwise, which proves (21). $\square$

Proof of Corollary 2. The stated condition makes the right-hand side of (21) at most ε . If $\eta \asymp T^{-1/2}$, $p_{\min}$ is constant, and the costs and proxy floor are fixed so that $\Lambda_{\text{on}} = O(1)$, then both excess terms in (21) are $O(T^{-1/2})$. $\square$

B Experimental Details

The experiments use the two ImageNet CSV files from the HW4 copy directory: ImageNet has $T = 50000$ items and ImageNetV2 has $T = 10000$ items. Each file contains a ground-truth label, a model prediction, and a confidence score. We treat the model prediction as the only cheap source and the ground-truth label as the expert. The proxy loss is $\hat{h} = 1 - \text{confidence}$, the cheap-source deployment cost is zero, and the expert cost is one. All experiments use 500 random trials. The scripts are `run_simple_offline_pac_experiment.py`, `run_simple_online_pac_experiment.py`, and `run_two_source_offline_pac_experiment.py`.

Offline implementation. For the fixed-price method, the candidate threshold grid contains the expert-only rule plus 201 evenly spaced confidence-loss thresholds between 0 and 1. Although Section 2.1 states the general finite-policy bound, this single-source experiment uses the sharper monotone threshold radius

$$\sqrt{\frac{\log(1/\alpha)}{2m}},$$

which is valid uniformly over all scalar thresholds by the same nested-class argument used in the original PAC baseline. For each trial, the calibration indices are sampled independently and uniformly from the target pool with replacement, with calibration size $m = 0.2T$. The confidence

level is $\alpha = 0.05$ and $\varepsilon \in \{0.05, 0.10, 0.15\}$. The original PAC baseline uses the same calibration sample and the same monotone radius, but chooses its threshold directly from the calibration mistake order instead of from the fixed grid. The full sweep over $\varepsilon \in \{0.05, 0.10, 0.15\}$ is reported in Table 4.

Dataset	Method	ε	Error	Violation	Budget saved
ImageNet	fixed-price monotone UCB	0.05	0.0372	0.0000	0.6296
ImageNet	original PAC	0.05	0.0377	0.0000	0.6323
ImageNet	fixed-price monotone UCB	0.10	0.0873	0.0000	0.7848
ImageNet	original PAC	0.10	0.0879	0.0000	0.7864
ImageNet	fixed-price monotone UCB	0.15	0.1372	0.0000	0.8851
ImageNet	original PAC	0.15	0.1380	0.0000	0.8866
ImageNetV2	fixed-price monotone UCB	0.05	0.0218	0.0000	0.3642
ImageNetV2	original PAC	0.05	0.0228	0.0000	0.3706
ImageNetV2	fixed-price monotone UCB	0.10	0.0718	0.0000	0.5457
ImageNetV2	original PAC	0.10	0.0726	0.0000	0.5479
ImageNetV2	fixed-price monotone UCB	0.15	0.1219	0.0000	0.6648
ImageNetV2	original PAC	0.15	0.1228	0.0000	0.6665

Table 4: Full offline sweep. Error, violation, and budget saved are averages over 500 trials. Rows are grouped by (dataset, ε) scenario and separated by light rules.

Two-source implementation. `run_two_source_offline_pac_experiment.py` constructs two complementary cheap sources from each base ImageNet table using the parity-of-true-class partition described in Section 4. The construction is seeded separately from the trial sampler so the synthesized pool is fixed across trials; defaults are `--synth-seed 20260526` and `--low-conf-max 0.2`. The calibration grid contains the expert-only policy plus 201 evenly spaced thresholds on $\min_j \hat{h}_{t,j}$, the calibration size is $m = 0.2T$, and $\alpha = 0.05$. The LP method uses the union-bound radius $\sqrt{\log(|\Lambda|/\alpha)/(2m)}$; the single-source baselines use the monotone radius $\sqrt{\log(1/\alpha)/(2m)}$; the Bonferroni-corrected best-of-singles baseline runs both single-source baselines at $\alpha/2$ each and reports the one with larger empirical savings on the calibration sample. Constructed-pool statistics are written to `results/two_source_offline_pac/two_source_offline_pac_construction.csv`. The full sweep over $\varepsilon \in \{0.05, 0.10, 0.15\}$ for all four methods is in Table 5.

Online implementation. For each trial, the target pool is randomly permuted into a stream. The update uses $\eta = 1$, proxy floor $\rho = 0.01$, and $\lambda_1 = 1/\varepsilon$. The online routing rule uses the AI prediction when $\lambda_t \max(1 - \text{confidence}_t, \rho) < 1$ and otherwise defers to the expert. If the AI prediction is used, it is audited with probability $p \in \{0.1, 0.2, 0.5\}$. The hybrid intermittent update is $-\eta\varepsilon_0$ on expert deferrals, $\eta(\ell_t - \varepsilon_0)/p$ on audited AI decisions, and zero on unaudited AI decisions. Audited AI labels are corrected by the expert before release; unaudited AI labels are released as-is. The reported “AI error” is the counterfactual average loss before audit correction, and the reported final error is the realized loss after correction. As noted in Section 4, these runs set $\varepsilon_0 = \varepsilon$ rather than subtracting the finite-horizon slack from Corollary 2. The full sweep over $\varepsilon \in \{0.05, 0.10, 0.15\}$ and $p \in \{0.1, 0.2, 0.5\}$ is reported in Table 6.

Dataset	Method	ε	Error	Violation	Budget saved
ImageNet	LP two-source uniform UCB	0.05	0.0290	0.0000	0.5898
ImageNet	Best-of-singles (Bonferroni)	0.05	0.0372	0.0000	0.3782
ImageNet	Source A monotone PAC	0.05	0.0378	0.0000	0.3743
ImageNet	Source B monotone PAC	0.05	0.0378	0.0000	0.3809
ImageNet	LP two-source uniform UCB	0.10	0.0793	0.0000	0.7645
ImageNet	Best-of-singles (Bonferroni)	0.10	0.0879	0.0000	0.4746
ImageNet	Source A monotone PAC	0.10	0.0878	0.0000	0.4715
ImageNet	Source B monotone PAC	0.10	0.0877	0.0000	0.4754
ImageNet	LP two-source uniform UCB	0.15	0.1290	0.0000	0.8705
ImageNet	Best-of-singles (Bonferroni)	0.15	0.1383	0.0000	0.5296
ImageNet	Source A monotone PAC	0.15	0.1377	0.0000	0.5273
ImageNet	Source B monotone PAC	0.15	0.1379	0.0000	0.5299
ImageNetV2	LP two-source uniform UCB	0.05	0.0019	0.0000	0.0936
ImageNetV2	Best-of-singles (Bonferroni)	0.05	0.0210	0.0000	0.2331
ImageNetV2	Source A monotone PAC	0.05	0.0227	0.0000	0.2254
ImageNetV2	Source B monotone PAC	0.05	0.0230	0.0000	0.2424
ImageNetV2	LP two-source uniform UCB	0.10	0.0534	0.0000	0.4923
ImageNetV2	Best-of-singles (Bonferroni)	0.10	0.0720	0.0000	0.3587
ImageNetV2	Source A monotone PAC	0.10	0.0727	0.0000	0.3470
ImageNetV2	Source B monotone PAC	0.10	0.0730	0.0000	0.3629
ImageNetV2	LP two-source uniform UCB	0.15	0.1036	0.0000	0.6250
ImageNetV2	Best-of-singles (Bonferroni)	0.15	0.1226	0.0000	0.4389
ImageNetV2	Source A monotone PAC	0.15	0.1234	0.0000	0.4287
ImageNetV2	Source B monotone PAC	0.15	0.1226	0.0000	0.4412

Table 5: Full two-source sweep. Error, violation, and budget saved are averages over 500 trials of calibration sampling. Rows are grouped by (dataset, ε) scenario and separated by light rules; the largest budget saved in each scenario is shown in bold.

Dataset	ε	p	Final error	AI error	Violation	Budget saved
ImageNet	0.05	0.1	0.0436	0.0485	0.0100	0.5738
ImageNet	0.05	0.2	0.0394	0.0493	0.0000	0.5244
ImageNet	0.05	0.5	0.0249	0.0497	0.0000	0.3344
ImageNet	0.10	0.1	0.0780	0.0866	0.0000	0.6629
ImageNet	0.10	0.2	0.0737	0.0921	0.0000	0.6115
ImageNet	0.10	0.5	0.0490	0.0981	0.0000	0.3952
ImageNet	0.15	0.1	0.1017	0.1130	0.0000	0.7159
ImageNet	0.15	0.2	0.0979	0.1223	0.0000	0.6626
ImageNet	0.15	0.5	0.0683	0.1367	0.0000	0.4330
ImageNetV2	0.05	0.1	0.0445	0.0494	0.1840	0.4140
ImageNetV2	0.05	0.2	0.0393	0.0491	0.0100	0.3750
ImageNetV2	0.05	0.5	0.0244	0.0489	0.0000	0.2372
ImageNetV2	0.10	0.1	0.0863	0.0958	0.0400	0.5091
ImageNetV2	0.10	0.2	0.0783	0.0979	0.0000	0.4681
ImageNetV2	0.10	0.5	0.0499	0.0996	0.0000	0.3010
ImageNetV2	0.15	0.1	0.1197	0.1331	0.0000	0.5731
ImageNetV2	0.15	0.2	0.1126	0.1407	0.0000	0.5330
ImageNetV2	0.15	0.5	0.0739	0.1479	0.0000	0.3465

Table 6: Full online sweep. Final error, AI error, and budget saved are averages over 500 trials. Rows are grouped by (dataset, ε) scenario and separated by light rules.

C Alternative Confidence Bounds

The Hoeffding bound (7) is simple but may be conservative. Any valid one-sided mean upper confidence bound can be substituted. For example, for a fixed λ , an empirical Bernstein bound gives

$$U_m^{\text{EB}}(\lambda; \alpha) = \hat{L}_m(\lambda) + \sqrt{\frac{2\hat{V}_m(\lambda) \log(2/\alpha)}{m}} + \frac{7 \log(2/\alpha)}{3(m-1)}, \quad (25)$$

where $\hat{V}_m(\lambda)$ is the sample variance of $H_{I_s}(\lambda)$. For a finite grid under unequal costs (Remark ??), replace α by $\alpha/|\Lambda|$; under equal costs the monotone single-price bound needs no such correction. Betting-based confidence sequences can also be used if the calibration stream is reused while expanding the candidate price set.

D Not Very Meaningful Results

The following observation is included only to show that a continuum of prices can be reduced to a finite policy class on a fixed finite dataset. It is relevant only under unequal cheap-source costs, where source selection varies with λ ; under the equal-cost setting of the main text Theorem 1 already certifies all prices without it. It is not a main contribution: the resulting confidence penalty may be larger than that of a deliberately chosen price grid, and it does not remove the uniform-certification cost.

D.1 Continuous Prices and Breakpoint Enumeration

For each item t , set $\hat{h}_t(E) = 0$ and $\hat{h}_t(j) = \hat{h}_{t,j}$. The action selected by a fixed price is the lower envelope of the $k + 1$ lines

$$c_t(a) + \lambda \hat{h}_t(a), \quad a \in \{E, 1, \dots, k\}.$$

With a deterministic tie-breaking rule, the action can change only when two of these lines cross.

Corollary 3 (Uniform certification over all prices). *Fix a bounded interval $[0, \lambda_{\max}]$ and include the expert-only policy as an additional candidate. Let Π_λ be the fixed-price policy induced by (??)–(9) on the union of the target and calibration items. The number of distinct induced policies over $\lambda \in [0, \lambda_{\max}]$ is at most*

$$N_{\text{br}} \leq 1 + 2(T + m) \binom{k+1}{2}. \quad (26)$$

Therefore the uniform guarantee holds over all $\lambda \in [0, \lambda_{\max}]$ after replacing $|\Lambda|$ in (??) by N_{br} . Under equal cheap-source costs this enumeration is unnecessary: source selection does not depend on λ , L_T is monotone, and Theorem 1 already certifies the whole continuum with the single-price bound (7).

Proof. For a fixed item, the selected action is constant between consecutive crossing points of pairs of lines $c_t(a) + \lambda \hat{h}_t(a)$. There are at most $\binom{k+1}{2}$ pairwise crossings per item. Across $T + m$ target and calibration items, there are therefore at most

$$B = (T + m) \binom{k+1}{2}$$

crossing values, counting multiplicity. These crossing values partition $[0, \lambda_{\max}]$ into at most $B + 1$ open intervals, and there are at most B exact crossing points. Within each open interval the entire vector of decisions on target and calibration items is fixed. At an exact crossing point, deterministic tie-breaking determines a single additional policy, which may or may not coincide with one of the neighboring interval policies. Thus the number of distinct induced policies is at most $2B + 1$. Applying Theorem 1 to one representative price from each induced policy gives the result. $\square$

D.2 Explicit Cost Accounting

This accounting is useful for implementation, but it is not central to the statistical guarantee. The main distinction is between deployment cost, development-label cost, and audit or correction cost.

Deployment labeling cost. For a fixed offline price λ , the average deployment cost on the target pool is exactly $C_T(\lambda)$ from (11), that is, $\frac{1}{T} \sum_{t=1}^T [D_t^\lambda c_{t,E} + (1 - D_t^\lambda) c_{t,j^\lambda}]$. This is the cost optimized in (8). If cheap predictions have already been precomputed at no marginal deployment cost, set $c_{t,j} = 0$. If calling a model has a per-token, per-image, or per-query cost, include that cost in $c_{t,j}$.

Offline development cost. If m_{train} expert labels are used to train $\hat{h}$ and m_{cal} expert labels are used to certify the fixed price, the development cost is

$$C^{\text{dev}} = \sum_{i \in I_{\text{train}}} c_{i,E} + \sum_{i \in I_{\text{cal}}} c_{i,E} + \text{cheap-source costs incurred on development items.} \quad (27)$$

This cost may be treated as sunk if the labeled development data already exist, but it should be included when comparing end-to-end data-curation pipelines.

Audit-or-correction cost. In the offline setting, calibration labels play the role that audits play online. If calibration items are part of the target pool and are corrected to expert labels in the released dataset, then there is no duplicate expert cost for those items. The total realized end-to-end cost can be written as

$$C_T^{\text{total}}(\lambda) = \frac{1}{T} \sum_{t=1}^T \left[(1 - D_t^\lambda) c_{t,j_t^\lambda} + \mathbf{1}\{D_t^\lambda \text{ or } t \in I_{\text{cal}}\} c_{t,E} \right] + C^{\text{train}}/T, \quad (28)$$

where C^{train} denotes expert-labeling cost used only for proxy training and not for final target labels. This formula separates expert labels used for final deferral from expert labels used for statistical certification.

Online audit cost. For Algorithm 2, the conditional expected round- t cost under audit-and-correct is

$$\mathbb{E}[C_t \mid \mathcal{G}_t] = D_t c_{t,E} + (1 - D_t) (c_{t,j_t^\star} + p_t c_{t,E}).$$

This identity isolates the extra expert cost paid to obtain online feedback; it is not a cost-regret guarantee.

References

- [1] Emmanuel J. Candès, Andrew Ilyas, and Tijana Zrnic. Probably approximately correct labels. *arXiv preprint arXiv:2506.10908*, 2025.
- [2] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.
- [3] Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511*, 2021.
- [4] Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems*, 2021.
- [5] Haosen Ge, Hamsa Bastani, and Osbert Bastani. Stochastic online conformal prediction with semi-bandit feedback. *arXiv preprint arXiv:2405.13268*, 2024.
- [6] Bowen Wang, Matteo Zecchin, and Osvaldo Simeone. Mirror online conformal prediction with intermittent feedback. *arXiv preprint arXiv:2503.10345*, 2025.

- [7] Junyoung Yang, Kyungmin Kim, and Sangdon Park. Online conformal prediction with adversarial semi-bandit feedback via regret minimization. *arXiv preprint arXiv:2604.17984*, 2026.
- [8] David Madras, Toniann Pitassi, and Richard Zemel. Predict responsibly: Improving fairness and accuracy by learning to defer. In *Advances in Neural Information Processing Systems*, 2018.
- [9] Hussein Mozannar and David Sontag. Consistent estimators for learning to defer to an expert. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 7076–7087, 2020.
- [10] Anqi Mao, Christopher Mohri, Mehryar Mohri, and Yutao Zhong. Two-stage learning to defer with multiple experts. In *Advances in Neural Information Processing Systems*, 2023.
- [11] Jean V. Alves, Diogo Leitão, Sérgio Jesus, Marco O. P. Sampaio, Javier Liébana, Pedro Saleiro, Mário A. T. Figueiredo, and Pedro Bizarro. Cost-sensitive learning to defer to multiple experts with workload constraints. *Transactions on Machine Learning Research*, 2024.